two criteria: the law must be satisfied by group division, and must fail for magmas that are not Ward quasigroups.

10.2. **Basic constraints.** There are 298 012 537 laws of order up to 8, and running an ATP on all of them is too slow, so one needs efficient ways to filter them beforehand. Let us begin with restrictions on the shape of any law equivalent to $\mathrm{E_{HN}}$. The law must take the form $\mathrm{x} \simeq \dots$ as otherwise it would be satisfied by the constant operation on any set. The law must be satisfied when evaluated with all variables set to the same element (say, 1) in the Ward quasigroup $\mathbb{Z}$ equipped with subtraction. In particular the law must have even order. This reduces from 3470 shapes of order up to 8, down to just 548 shapes.

Next come some restrictions on the variables. The right-hand side must not start nor end with the variable x, as otherwise the projection operations $x \diamond y \coloneqq x$, $x \diamond y \coloneqq y$ respectively would satisfy the law. The law must have at least three variables: otherwise it is satisfied by division in any diassociative loop (such as a Moufang loop), namely a quasigroup with identity element in which every 2-generated submagma is a group. Each variable must appear an even number of times, so that the law holds in Boolean groups (abelian groups of exponent 2). These basic constraints leave 54, 9000, and 1 841 910 candidate laws of orders 4, 6, and 8, respectively, which can be efficiently enumerated since the conditions so far constrain separately the shape and list of variables (refer to Appendix A for the relevant definitions).

Imposing further that the law is satisfied by division in a free non-abelian group (with one generator per variable) reduces these numbers of laws to 0, 59, and 5692 at these same orders. All of the laws coming out of these filters are consequences of $\mathrm{E_{HN}}$; accordingly, one must determine which of these candidates imply said law.

10.3. **Using automated theorem provers.** We repeatedly whittled down the list of candidates by accumulating a collection of finite countermodels, namely magmas that satisfy a candidate law while violating one of the laws E11, E40 and E3737 characterizing Ward quasigroups. Automated searches of small magmas (of size up to 8) with *Mace4* or *Vampire* gave many countermodels (of which 12 are enough). A second source was that of linear models $x \diamond y \coloneqq ax + by$ on $\mathbb{Z}/n\mathbb{Z}$ with $(a, b) \neq (1, n-1)$: the largest one we used is $x \diamond y \coloneqq 261x + 33y \bmod 307$ to rule out the candidate law E68185620, $\mathrm{x} \simeq (\mathrm{y} \diamond \mathrm{y}) \diamond (\mathrm{y} \diamond ((\mathrm{x} \diamond (\mathrm{z} \diamond \mathrm{y})) \diamond ((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{z})))$. Finally, we introduced some models that are "almost" Ward quasigroups: the 7-element smallest non-associative inverse loop (equipped with division), the 10-element smallest non-associative Steiner loop (commutative loop in which divisions coincide with multiplication), and the 16-element Moufang loop of unit octonions over $\mathbb{Z}$.

These steps eliminated all candidates of order less than 8,[23] and left only 213 laws of order 8 that could be equivalent to $\mathrm{E_{HN}}$. These laws come in 31 families consisting of a "parent"

[23]In particular we recover the nonexistence of laws of order 6 characterizing Ward quasigroups, already announced by McCune and Kinyon [46].

5-variable law and some specializations with pairs of variables being identified. The lowest-numbered law in this list is McCune's law

$$\mathrm{x} \simeq \mathrm{y} \diamond \Big(\big(((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x}) \diamond \mathrm{z}\big) \diamond \big(((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{y}) \diamond \mathrm{z}\big)\Big) \text{ (E42302852)},$$

which is in the same family as the Higman–Neumann law. Another common feature is that all 213 candidate laws include at least one subexpression of the form $v \diamond v$ for some variable v.

For 179 candidate laws E, we showed the implication $\mathrm{E} \models \mathrm{E}_{\mathrm{HN}}$ using the ATP *Prover9*. For equations of this order, the ATP computation times increase significantly compared to order-4 laws, with some proofs taking 20 times longer than checking with *Prover9* all 8 178 279 positive implications of the main project. The choices of parameters bounding the ATP search (such as the parameter `max_weight` limiting clause complexity in *Prover9*) were particularly crucial, with different values being optimal in different proofs. Another important speed-up was obtained by seeking proofs of a simple property such as E11, E40, E3737, or injectivity/surjectivity of left or right multiplications, then seeking proofs that the candidate law together with that property implies some other property, and so on, until proving all three laws characterizing Ward quasigroups. The reverse approach also proved useful, namely finding which property would allow the proof to succeed, then seeking a proof of that property from the candidate law. The law E102744082 was a particularly difficult instance: together with injectivity of right multiplications it easily implies the Higman–Neumann law, but the proof that E102744082 does imply injectivity took several hours to obtain in a sweeping search with general parameters; an optimized choice of *Prover9* options trims this time down to 0.2 seconds. The two characterizations E42302946 and E89176740 of division in groups deserve particular mention for being nicely expressed in terms of the right-cubing map $C(x) \coloneqq (x \diamond x) \diamond x$:

$$\mathrm{x} \simeq \mathrm{y} \diamond ((C(\mathrm{x}) \diamond \mathrm{z}) \diamond (C(\mathrm{y}) \diamond \mathrm{z})), \qquad \mathrm{x} \simeq C(\mathrm{y}) \diamond ((C(\mathrm{x}) \diamond \mathrm{z}) \diamond (\mathrm{y} \diamond \mathrm{z})).$$

Among the 34 remaining candidates, we showed the finite implication $\mathrm{E} \models_{\mathrm{fin}} \mathrm{E}_{\mathrm{HN}}$ for 21 laws E, which means that the law E characterizes Ward quasigroups among finite magmas. Let us illustrate the proof technique for $\mathrm{x} \simeq (\mathrm{y} \diamond \mathrm{y}) \diamond (\mathrm{y} \diamond ((\mathrm{x} \diamond \mathrm{z}) \diamond (((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{y}) \diamond \mathrm{z})))$ (E67953597). In a finite magma, one gets $x = L_{y \diamond y} \circ L_y \circ f_{y,z}(x)$ in terms of the function $f_{y,z} \colon x \mapsto (x \diamond z) \diamond (((x \diamond x) \diamond y) \diamond z)$. Finiteness implies that the composition of several functions can only be a bijection if all of them are bijections, thus left multiplications are bijective. By selecting $y = L^{-1}_{x \diamond x}(w)$ one gets that $(x \diamond z) \diamond (w \diamond z)$ equals the z-independent expression $L^{-1}_y \circ L^{-1}_{y \diamond y}(x)$. Taking $w = x$ yields that the square of $L_x(z)$ is z-independent, hence (by surjectivity of L_x) all squares are equal. A routine ATP run then concludes. While the resulting proofs of finite implications are relatively short and have been successfully ported to *Lean*, our automated search involved thousands of *Vampire* runs. Indeed, rather than the condition that a bijective composition implies bijectivity of its constituents, we had to use the more concrete property that injectivity is equivalent to surjectivity for various collections of specific functions $f \colon M \to M$ such as left or right multiplications, cubing, etc., with a brute-force search over which functions to include in a given run.

The remaining 13 candidates have proven to be quite resistant to both proof and counter-model attacks with a wide range of parameter options, target clauses, additional clauses, and given time (up to 10 hours for each candidate and given experiment). We have shown

with *Prover9* that a model of any of the remaining candidates E that has a right-identity element, or that satisfies the right-unit squares law E11 or unipotence law E40, is a Ward quasigroup. As such, any putative countermodel to the implication $E \models E_{HN}$ must be far from being a Ward quasigroup, in the sense that it must violate these laws.[24] In particular it cannot be a loop; this discards, e.g., the 32-element loop of unit sedenions over $\mathbb{Z}$ as a potential countermodel.

In summary, out of the 298 012 537 laws of order up to 8, we found 179 laws characterizing Ward quasigroups, 21 characterizing them among finite magmas but perhaps not infinite ones, and 13 candidates for which we have neither a counterexample nor a proof even for finite magmas. No efforts have been made yet to find infinite countermodels with the techniques developed in the main project. The results of this section[25] have not been formalized in *Lean* yet. More details and further progress on this side project will be published elsewhere.

11. AI and Machine Learning contributions

As discussed in Section 7, the ETP made extensive use of automated theorem provers in completing the primary goal of determining and then formalizing all the implications between the specified equational laws. In contrast, we were only able to utilize modern large language models (LLMs) in a fairly limited fashion. Such models were useful in writing initial code for graphical user interfaces that we discuss further in Section 12, as well as performing some code autocompletion (using tools such as *GitHub Copilot*) when formalizing an informal proof in *Lean*. In one instance, *ChatGPT* was used[26] to guess a complete rewriting system for the law $x \diamond y \simeq x \diamond ((y \diamond y) \diamond z)$ (E3523) which could then be formally verified, thus resolving all implications from this equation. However, in most of the difficult implications that resisted automated approaches, we found that LLMs did not provide useful suggestions beyond what the human participants could already propose.

On the other hand, we found that machine learning (ML) methods showed some promise of being able to heuristically predict the truth value of portions of the implication graph; we shall now discuss a convolutional neural network approach.[27]

Convolutional neural network model for the implication graph. To model the implication graph, we used a convolutional neural network (CNN). For each pair of equations (p, q), the input of the CNN consisted of a character-level tokenization (no vectorization) of

[24] In fact, we find that countermodels must violate many laws satisfied by group division: E11, E40, E823, E835, E842, E846, E1049, E1637, E1673, E1718, E1835, E1876, E3282, E3323, E3662, E3665, E3677, E3684, E3721, E3729, E3737, E3761, E3823, E3870, E3891, E3943, E4270, E4590 of order up to 4, and many more of higher order, including the associative right division law E912704, $x \diamond (y \diamond (z \diamond w)) \simeq (x \diamond (w \diamond z)) \diamond y$.

[25] `https://github.com/teorth/equational_theories/blob/main/data/Higman-Neumann.json`

[26] `https://chatgpt.com/share/670ce7db-8a44-800d-a5dc-8462c12eca3b`

[27] For some discussion of other machine learning experiments performed during the Equational Theories project, see `https://leanprover.zulipchat.com/#narrow/channel/458659-Equational/topic/Machine.20learning.2C.20first.20results` for a (vectorized) transformer neural network approach, and see `https://leanprover.zulipchat.com/#narrow/channel/458659-Equational/topic/Graph.20ML.3A.20Directed.20link.20prediction.20on.20the.20implication.20graph` for directed link prediction on the implication graph using Graph Neural Network (GNN) autoencoders.